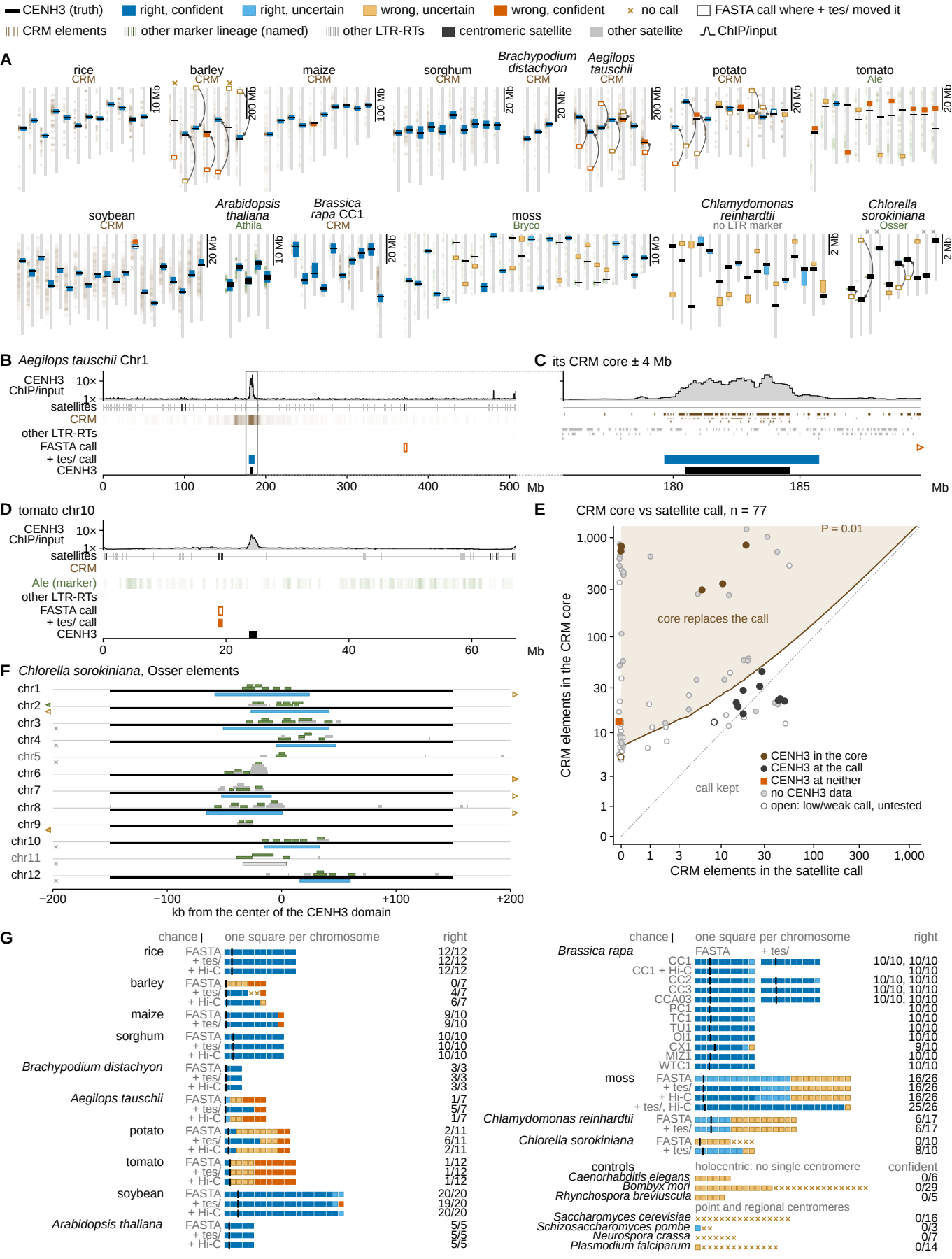

CENH3 (truth)
right, confident
right, uncertain
wrong, uncertain
wrong, confident
no call
FASTA call where + tes/ moved it
CRM elements
other marker lineage (named)
other LTR-RTs
centromeric satellite
other satellite
ChIP/input
A
rice
CRM
barley
CRM
maize
CRM
sorghum
CRM
Brachypodium distachyon
CRM
Aegilops tauschii
CRM
potato
CRM
tomato
Ale
soybean
CRM
Arabidopsis thaliana
Athila
Brassica rapa CC1
CRM
moss
Bryco
Chlamydomonas reinhardtii
no LTR marker
Chlorella sorokiniana
Osser
B Aegilops tauschii Chr1
CENH3 ChIP/input
satellites
CRM
other LTR-RTs
FASTA call
+ tes/ call
CENH3
Mb
C its CRM core ± 4 Mb
D tomato chr10
Ale (marker)
E
CRM core vs satellite call, n = 77
P = 0.01
core replaces the call
call kept
CENH3 in the core
CENH3 at the call
CENH3 at neither
no CENH3 data
open: low/weak call, untested
CRM elements in the CRM core
CRM elements in the satellite call
F Chlorella sorokiniana, Osser elements
kb from the center of the CENH3 domain
G
chance
one square per chromosome
right
rice 12/12 12/12 12/12
barley 0/7 4/7 6/7
maize 9/10 9/10
sorghum 10/10 10/10 10/10
Brachypodium distachyon 3/3 3/3 3/3
Aegilops tauschii 1/7 5/7 1/7
potato 2/11 6/11 2/11
tomato 1/12 1/12 1/12
soybean 20/20 19/20 20/20
Arabidopsis thaliana 5/5 5/5 5/5
Brassica rapa
CC1 10/10, 10/10
CC1 + Hi-C 10/10
CC2 10/10, 10/10
CC3 10/10, 10/10
CCA03 10/10, 10/10
PC1 10/10
TC1 10/10
TU1 10/10
OI1 10/10
CX1 9/10
MIZ1 10/10
WTC1 10/10
moss 16/26 16/26 16/26 25/26
Chlamydomonas reinhardtii 6/17 6/17
Chlorella sorokiniana 0/10 8/10
controls
holocentric: no single centromere
Caenorhabditis elegans 0/6
Bombyx mori 0/29
Rhynchospora breviuscula 0/5
point and regional centromeres
Saccharomyces cerevisiae 0/16
Schizosaccharomyces pombe 0/3
Neurospora crassa 0/7
Plasmodium falciparum 0/14

Plant centromeres can be found from the genome sequence alone where they sit on satellite arrays, and from clusters of one retrotransposon lineage where they do not; counting that lineage's insertions decides between the two. Centromeres marked by neither stay unfound.

**A** 160 chromosomes of 14 plant and algal genomes, each drawn to scale, with the measured CENH3 domain (black). Calls from the sequence alone mostly land on CENH3; where they miss (barley, potato, *Aegilops tauschii*, *Chlorella*), the retrotransposon line moves many of them onto it (arrows). Tomato stays wrong.

**B, C** An *Aegilops tauschii* chromosome whose CENH3 sits on a dense cluster of CRM retrotransposons with no satellite; the sequence-only call lies on a satellite array far away. Up close, hundreds of CRM insertions fill the CENH3 domain.

**D** A tomato chromosome whose CENH3 lies on DNA holding neither satellite nor CRM: the call follows a satellite elsewhere and is wrong.

**E** When a CRM cluster and a satellite call disagree, the one holding significantly more CRM insertions is taken; CENH3 sat on the chosen side in 14 of 15 tested cases with CENH3 data, and at neither in the last.

**F** In the green alga *Chlorella*, each centromere is a handful of insertions of one Copia lineage (Osser), and nearly every Osser copy in the genome sits in a centromere.

**G** Every chromosome as a square: far above chance from the sequence alone in satellite genomes, rescued by retrotransposons in four more and, in moss, by retrotransposons and Hi-C together; still wrong in tomato. No confident call in control genomes that are holocentric or have point or regional centromeres.

**Terms.** Calls are from one rerun of cenfuse 0.10.1 on every benchmark arm (2026-11-04), scored with the rules fixed before each benchmark was first run; E adds conflicts logged while the CRM rule was developed. *Truth*: CENH3 domains from ChIP-seq (CUT&Tag in *Chlamydomonas*) of the same line, called with one fixed rule (100-kb bins, ChIP over input, the half-maximum run around the peak, at least 4-fold): *B. rapa* (9 accessions), soybean X26, sorghum BTx623 (ChIP of line V9), *Ae. tauschii* AL8/78, *Chlorella sorokiniana* NS4-2, tomato Heinz 1706, maize Mo17, barley MorexV3, *Brachypodium* Bd21 (3 of 5 chromosomes reach 4×), *Chlamydomonas* UL-1690, *Arabidopsis* Col-0; published CENH3 domains for rice (Shang et al. 2023), potato (Pham et al. 2020), moss (*Physcomitrium patens*, Bi et al. 2024) and *B. rapa* TU1 and CX1 (ChIP too weak). *Right*: the call lies within 1 Mb of the domain (moss and algae 100 kb, barley 8 Mb, *Arabidopsis* overlap; rice, potato and barley with Hi-C, 4 Hi-C bins). *Confident* = cenfuse confidence high or medium; *uncertain* = low. *Chance*: the probability that a call of the same width placed at random is right, summed over chromosomes. *FASTA*: the assembly alone; + *tes/*: with LTR retrotransposons (LTRquest); + *Hi-C*. CRM: the centromeric retrotransposon lineage of Ty3/Gypsy chromoviruses. Every call is shown; uncertain calls and misses are shown on purpose.

**A** Gray bars: chromosomes, each genome on its own scale (bar = Mb); black: CENH3 domain; box: the + tes/ call, colored by outcome; open box and arrow: the FASTA call where + tes/ moved it, edge colored by its outcome; ×: no call (gray: FASTA only). Ticks left of each bar: every element of the genome's centromeric LTR lineage, CRM (brown) or the lineage cenfuse follows where CRM does not concentrate (green, named: Bryco in moss, Osser in *Chlorella*, Athila in *Arabidopsis*, Ale in tomato); *Chlamydomonas* has none (its centromeres are LINE clusters). All 160 chromosomes of the genomes run with tes/ (*B. rapa* CC1 for its 4 accessions; the other 10 in G). + tes/ changed 30 calls: 20 wrong → right, 1 right → wrong (soybean chr13), 7 wrong → still wrong, 2 right → right. Over the same chromosomes, FASTA 125/190 right (17 confidently wrong), + tes/ 144/190 (14 confidently wrong; chance 22.9).

**B, C** *Aegilops tauschii* AL8/78 Chr1, one bp axis; C (same track rows) is the CRM core ± 4 Mb (175.7-189.8 Mb), every element to size, overlapping copies stacked. Tracks: CENH3 ChIP/input (100-kb bins, 3-bin mean: the truth rule's own track, from 195,266,471 / 100,984,317 mapped ChIP / input fragments; maximum 13.2×); satellite arrays (dark: families the FASTA run ranked centromeric; dispersed families omitted); CRM and all other LTR retrotransposons, every element at its position; FASTA call; + tes/ call; CENH3. The FASTA call (370.3-371.0 Mb, high) is a 369-bp satellite array holding 0 CRM elements. The CRM core (179.7-185.8 Mb, 6.1 Mb wide) holds 300 of Chr1's 784 CRM elements (38%), no satellite array and the whole CENH3 domain (180.5-184.6 Mb), whose enrichment peaks inside it; + tes/ calls it (medium). Chosen by rule: of the 12 chromosomes whose call + tes/ moved onto CENH3 by a CRM core (barley, potato, *Ae. tauschii*), the lower median of the move (187.9 Mb; upper median Chr4, 199 Mb).

**D** Tomato Heinz 1706 chr10, tracks as in B, with the Ale lineage cenfuse follows in tomato (green). Chosen by rule: the median call-to-CENH3 distance among tomato's 7 confidently wrong calls (1.9, 2.3, 3.5, 4.4, 11.3, 17.4, 41.8 Mb). The call (18.8-19.4 Mb, high, in FASTA and + tes/ alike) is on a 181-bp satellite, 4.4 Mb from CENH3 (23.8-25.0 Mb, 7.6×), whose domain holds 0 bp of satellite array. Tomato has 24 CRM elements in all, 2 of them inside its 12 CENH3 domains; the Ale core lies at 3.9-5.4 Mb. With 43.0 M Hi-C pairs the Hi-C test failed and no call changed (this chromosome's prior-free Hi-C peak: 15.5 Mb).

**E** One point per conflict between a decisive CRM core and a satellite call (77: 13 in benchmark genomes, the others in 28 polyploid and progenitor plant assemblies and in *Poa annua*, most without CENH3 data): calls away from the core logged while the rule was developed, satellite arrays beside the core (the benchmark's tes/ runs and a replay of 72 batch jobs, counted with the released engine's reading of each tes/ file) and the 6 calls the release kept against a core. Axes: CRM elements (midpoints) in the call and in the core, log scale. Line: P = 0.01 of the binomial test at ½ (shaded: the core replaces the call); open points: low or weak calls, which a core replaces untested; dotted: equal counts. With CENH3 data, 14 of 15 tested conflicts fell on the side CENH3 took (5 core, 9 call); in the other (*Solanum verrucosum* chr05) CENH3 is at neither. Uncertain calls with data, replaced untested: potato chr10, 5 vs 0, CENH3 in the core; soybean chr13, 13 vs 8, CENH3 at the call (counted on the FASTA-only call; the replaced call held 7). Not drawn: *Ae. tauschii* Chr1, Chr3 and Chr6, whose replaced calls the stored satellite searches do not keep.

**F** *Chlorella sorokiniana* NS4-2: each of its 12 chromosomes centered on its CENH3 domain (chr5 and chr11, gray: no domain ≥ 4×, centered on the published centromere), ± 200 kb; boxes: every LTR retrotransposon to size (green: Osser, a Copia lineage; nested copies stacked); triangles at the edge: an Osser copy or a call outside the window. 53 of the genome's 54 Osser copies lie in the published centromeres, 2-7 per centromere. FASTA: 0/10 right (6 uncertain, 4 no call); + tes/: 8/10, all uncertain (P $1.6 \times 10^{-6}$), the Osser cores calling them since the core floor counts 3 independent insertion sites; misses chr6, chr9.

**G** One square per chromosome, sorted by outcome; tick: chance; right/total at the right. FASTA alone: 194/260 right over all 14 species (chance 40.0, P $6.1 \times 10^{-119}$, 17 confidently wrong). *B. rapa* 109/110 (11 accessions) and with tes/ 40/40 (39 confident); soybean 20/20 → 19/20; *Ae. tauschii* 1/7 → 5/7 (P 0.068 → $2.4 \times 10^{-8}$); barley 0/7 → 4/7, with Hi-C 6/7 (arm fold); potato with Hi-C 2/11, with tes/ 6/11; *Chlorella* 0/10 → 8/10. Moss: 16/26 from the FASTA, all uncertain. With only uncertain calls to test it against, Hi-C agrees with 10-11 of 25, beyond chance but not most: it may only support, and raises 10 calls to medium. With tes/, 11 calls are confident; against them Hi-C agrees with 8-9 of 10-11, passes on all 26 chromosomes and moves 9 calls onto CENH3: 25/26, all confident (P $6.4 \times 10^{-32}$). Hi-C added to sorghum, *Brachypodium*, *Ae. tauschii*, soybean, *Arabidopsis* and *B. rapa* CC1 changed no outcome (soybean chr13 uncertain → confident; the test passed on 20/20 soybean, 10/10 CC1 and 2/5 *Arabidopsis* chromosomes, none elsewhere). Adding tes/ to Hi-C changes no count (rows not drawn): potato as + tes/; rice, barley (a different miss) and tomato as + Hi-C. Controls, FASTA alone (no chance tick: what counts is any confident call): holocentric *C. elegans*, *Bombyx mori* and *Rhynchospora breviuscula* 0/24/16/40 (confident/uncertain/none/total); point and regional centromeres of yeasts, *Neurospora* and *Plasmodium* 0/2/38/40 (the uncertain *S. pombe* call is right).

**Exceptions.** Soybean chr13: a right but uncertain satellite call is replaced by a CRM core 1.9 Mb from CENH3 (core 13 vs call 7 elements; uncertain calls are replaced without the test). *Ae. tauschii* Chr5 and Chr7: calls at the edge of the CRM core, 1.3 and 3.1 Mb from CENH3, stay wrong. Barley with tes/ but no Hi-C: 1 call confidently wrong, 2 with no call. Moss: Bryco, its centromeric lineage, may only flag (only CRM decides), so tes/ alone leaves 16/26, although its cores lie on CENH3 on 25 of 26 chromosomes; with Hi-C, Chr12 stays wrong. *Chlamydomonas* (6/17, all uncertain): its centromeres are LINE clusters, outside the LTR line. Tomato: a reported failure; neither Hi-C nor retrotransposons fix it. Sorghum truth is ChIP of another line (V9). Examples B and D were chosen by the stated median rules, not for looks; C is B's core.

Analysis:
asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/figures/benchmark_plants · Script:
/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/figures/benchmark_plants/benchmark_plants.py · 2026-11-04